

3D Reconstruction Method for Strawberry Plants Based on 3D Gaussian Splatting and Edge Detection

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Abstract—During the 3D reconstruction of strawberry plants, methods based on 3D Gaussian Splatting (3DGS) face significant challenges due to motion-induced image blur. Such blurring substantially reduces the feature matching accuracy in Structure from Motion (SfM) algorithms and compromises the reliability of camera pose estimation, thereby degrading the quality of subsequent 3DGS reconstruction. This ultimately manifests as geometric distortion and loss of texture details in the reconstructed models. The issue is particularly severe on the surface of strawberry fruits: under blurred image conditions, point cloud registration fails, resulting in the loss of high-frequency details in the high-density achene regions, which blurs seed contours and degrades reconstruction accuracy. To address this technical bottleneck, this study proposes an optimized reconstruction scheme integrated with 3DGS. By incorporating the Canny edge detection algorithm to filter high-quality input images, the accuracy of the reconstructed model is significantly improved. The optimized approach achieves remarkable results on the strawberry plant dataset: the average Peak Signal-to-Noise Ratio (PSNR) of the 3DGS model reaches 35.99, representing a 15.2% improvement over the baseline 3DGS. The morphology of high-density achenes on the fruit surface is clearly distinguishable, supporting the accurate monitoring of phenotypic parameters in strawberry plants.

Index Terms—3D Gaussian splatting, edge detection, 3D reconstruction

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I. INTRODUCTION

High-precision 3D reconstruction of strawberry plants holds significant application value in precision agriculture, providing digital support for critical tasks such as growth monitoring [1], phenotypic trait analysis [2], and fruit localization for autonomous harvesting machinery [3]. By accurately capturing detailed geometric features of plant organs—including fruit distribution patterns and leaf venation topology—the reconstructed models facilitate the quantification of phenotypic parameters that are difficult to obtain through conventional means, thereby establishing a crucial data foundation for varietal breeding and cultivation management decisions [4]. This application domain necessitates a reconstruction process that balances geometric fidelity with real-time interactivity to ensure visual authenticity on mobile and VR/AR platforms [5]. However, current technical solutions face considerable limitations: manual modeling relying on software like 3ds Max, Unity, or Blender and skilled operators is cost-prohibitive and often simplifies complex structures such as multi-level branches and leaves, resulting in geometric distortion [6], equipment-based scanning approaches, including LiDAR systems [7], require expensive specialized hardware and substantial computational resources for post-processing correction, meanwhile, traditional photogrammetry is constrained by its susceptibility to weak texture matching failures and varying lighting conditions, with computational and time costs escalating exponentially with increasing scene complexity [8].

These limitations are accelerating a paradigm shift from traditional geometric methods toward neural network-driven reconstruction techniques [9]. Such neural approaches lever-

age implicit continuous scene representations to model intricate structures, employ end-to-end optimization to suppress error propagation, and utilize transfer learning for cross-cultivar generalization—effectively transforming plants into high-fidelity computable models. Nevertheless, the performance of neural modeling remains highly dependent on input data quality. Motion blur or uneven lighting in source images can introduce artifacts into the implicit representation, substantially compromising the error resilience of end-to-end optimization [10].

To address the specific challenge of motion blur during in-field mobile image acquisition, this study introduces a dynamic frame filtering mechanism as an input optimization step. This approach enhances feature matching accuracy within the Structure from Motion (SfM) pipeline and improves camera pose estimation reliability, thereby curbing error propagation and providing high-precision 3D geometric representations for plant phenotyping analysis and intelligent agricultural operations.

II. RELATED WORK

Crop three-dimensional reconstruction serves as a crucial technical foundation for precision agriculture and digital plant phenotyping. Early approaches, primarily based on Structure from Motion-Multi-View Stereo (SfM-MVS), automatically estimated camera poses and reconstructed dense point clouds through feature matching across multi-view images, achieving high accuracy under favorable texture and lighting conditions. For instance, Peng et al. [11] proposed a binocular vision-based SfM method that captures multi-view images using a stereo camera and leverages feature matching combined with stereo vision to reconstruct 3D point cloud models, thereby extracting parameters like plant height and canopy width. Despite enabling dense reconstruction, SfM-MVS is computationally intensive and time-consuming, and its results often require extensive post-processing such as filtering, hole filling, and meshing to enhance usability [12].

To overcome the limitations of traditional methods in modeling efficiency and detail restoration, Neural Radiance Fields (NeRF) [13] have been introduced into crop 3D modeling in recent years. NeRF utilizes multi-layer perceptrons to map 3D coordinates and viewing directions to colors and densities, achieving an implicit continuous field representation. Hu et al. [14] applied NeRF to plant structure modeling, demonstrating high-quality representation of complex architectures and enhanced detail reconstruction. Similarly, Zheng et al. [15] validated its capability for highly faithful reconstruction of fruit geometry and phenotypic attributes in tomato plants. Although NeRF excels in visual quality, it faces practical bottlenecks including lengthy training times, slow rendering speeds, and poor adaptability to dynamic scenes [16].

Addressing these issues, Kerbl et al. [17] proposed 3D Gaussian Splatting (3DGS), a method that employs explicit point cloud representation and a differentiable Gaussian rendering mechanism. This approach increases reconstruction

speed by several orders of magnitude while maintaining quality comparable to NeRF [18]. A key advantage lies in its use of anisotropic Gaussian distributions to characterize spatial points, combined with fast occlusion modeling and GPU-accelerated rasterization, making it highly suitable for real-time interactive applications like VR/AR and digital twins [19]. Subsequent studies, such as Zhang et al.'s Wheat3DGS framework [20] for high-density spike modeling and Westra et al.'s method for tomato leaf area index estimation [21], demonstrate 3DGS's efficacy in preserving fine details and quantifying structural parameters. The evolution from SfM-MVS to NeRF and then to 3DGS signifies a paradigm shift in crop modeling—from geometry-driven approaches to neural representations, and further towards efficient implicit-explicit hybrid representations, offering a new solution for high-fidelity, interactive 3D reconstruction.

III. METHOD

This study proposes an optimized 3D reconstruction framework for strawberry plants. The framework operates through the following stages: first, a multi-view image acquisition system captures the plant data, and the Structure from Motion (SfM) technique is applied to generate an initial point cloud, which is subsequently converted into parameters for Gaussian ellipsoids. Then, during the preprocessing stage, an edge detection algorithm is employed to filter out blurred frames, thereby enhancing the quality of the input data. Finally, the parameters of the Gaussian ellipsoids are iteratively optimized via gradient descent and are projected and composited into a 2D image in a differentiable manner, leading to high-quality model rendering.

A. 3D Gaussian Splatting for Plant Reconstruction

The pipeline for the crop surface reconstruction method based on 3D Gaussian Splatting is illustrated in Fig. 1. The process begins with the acquisition of multi-view plant images through orbital photography. These images are then processed by Structure from Motion (SfM) to generate an initial sparse point cloud, which is subsequently converted into a set of 3D Gaussian kernels. Each kernel is parameterized by its position, orientation, scale, opacity, and spherical harmonic coefficients representing color.

To enhance spatial resolution, an adaptive densification process is applied, which employs splitting and cloning operations to refine geometric details and reinforce multi-view consistency. This step is crucial for improving the reconstruction accuracy of intricate structures like leaves and stems. The Gaussian parameters are then iteratively optimized through a gradient descent strategy to minimize the photometric loss between rendered and ground truth images.

In the final differentiable rendering stage, the optimized 3D Gaussians are projected onto the 2D image plane, sorted by depth, and composited pixel-by-pixel using a fast tile-based rasterizer. This efficient process ultimately synthesizes a high-fidelity novel view of the plant.

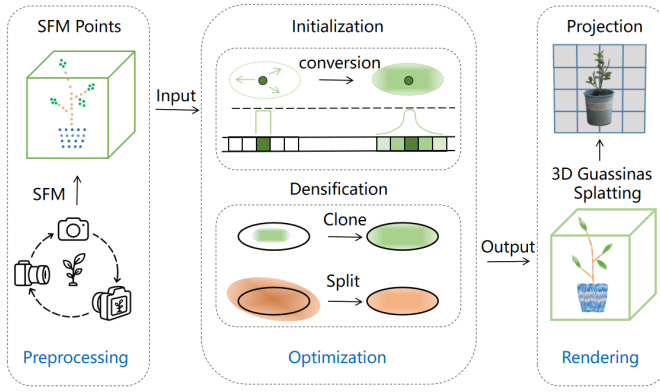


Fig. 1: 3D Gaussian splatting-based crop surface reconstruction method.

B. Canny Edge Detection for Dataset Optimization

To enhance modeling accuracy, this study introduces a dataset optimization method based on the Canny edge detection algorithm. The reconstruction quality of 3D Gaussian Splatting is highly dependent on input image quality. In practice, the input images are typically extracted from videos captured by an orbiting camera, a process susceptible to defocus and motion blur. Such image degradation adversely affects the robustness of the Structure-from-Motion (SfM) initialization and the fidelity of the resulting 3D model. To mitigate this issue, the Canny edge detector is employed to filter out frames that are blurry or lack sufficient edge information. This algorithm effectively assesses image clarity by quantifying edge intensity. Its core principle involves smoothing the image with a Gaussian filter to suppress noise before extracting salient edges. The Gaussian filter is defined by the following equation:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}. \quad (1)$$

$G(x, y)$ denotes the value of the Gaussian kernel at position (x, y) , and σ represents the standard deviation of the kernel, controlling the smoothness degree. Subsequently, the gradient magnitude at each pixel is computed using the Sobel operator. The gradient magnitude G is given by:

$$G = \sqrt{G_x^2 + G_y^2}, \quad (2)$$

where G_x and G_y represent the horizontal and vertical gradient components, respectively. To refine the edge map, non-maximum suppression is applied, which preserves only the local maxima in the gradient magnitude, thereby thinning the edges and suppressing non-edge pixels. Finally, dual-threshold detection is performed: pixels with gradient magnitudes above the high threshold are classified as strong edges and retained directly, while those between the high and low thresholds are considered weak edges and are retained only if they are connected to strong edges, ensuring robust edge connectivity and noise immunity.

IV. EXPERIMENT

The experiment employs an image sharpness weighted fusion model to evaluate the performance of edge detection algorithms in identifying motion-blurred and defocused images. Quantitative assessment of the 3D reconstruction framework's performance was conducted using image quality evaluation metrics. All experiments were performed on a unified RTX A6000 platform with parameters adhering to the original literature specifications. Image acquisition utilized an IMX766 sensor with default configurations, maintaining a frame overlap rate exceeding 60% and a baseline angle below 8 degrees under natural greenhouse illumination.

A. Image Sharpness Assessment

This study utilizes edge density and edge continuity, derived from edge detection algorithms, as core metrics for evaluating image sharpness. These complementary metrics effectively distinguish between different blur types. Edge density, which quantifies the richness of image details, is particularly sensitive to defocus blur, as optical defocus directly attenuates high-frequency information and reduces detectable edges. Conversely, edge continuity, which characterizes the structural integrity of edges, is more effective in identifying motion-induced blur, where camera shake or object movement disrupts edge coherence. Building upon this complementary relationship, we construct a weighted fusion model that adaptively integrates both metrics. By jointly considering detail richness and structural integrity, the model achieves a more accurate and robust assessment of image sharpness, overcoming the limitations of single-metric approaches. Edge density D , defined as the proportion of edge pixels, is calculated as:

$$D = \frac{N_{edge}}{N_{total}}, \quad (3)$$

where N_{edge} denotes the number of edge pixels, and N_{total} represents the total number of pixels in the image.

Edge continuity C , which measures edge connectivity, is computed as:

$$C = \log \left(1 + \frac{N_{edge}}{N_{boundary}} 2 \right), \quad (4)$$

where $N_{boundary}$ is the number of boundary pixels obtained via the dilation-erosion difference method.

The final sharpness score S is given by the weighted fusion model:

$$S = \min(D_{norm} \cdot w_d + C_{norm} \cdot w_c + B, 100), \quad (5)$$

where D_{norm} and C_{norm} are the normalized values of edge density and continuity respectively, w_d and w_c are corresponding weighting coefficients, and B is a continuity reward term.

B. Reconstruction Quality Evaluation

To quantitatively evaluate the performance of the 3D Gaussian reconstruction model, this study employs a comprehensive set of metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS).

PSNR is a fundamental objective image quality metric that quantifies reconstruction fidelity through the mean squared error (MSE) between the original and reconstructed images. A higher PSNR value indicates lower distortion and superior reconstruction quality. It is defined as:

$$\text{PSNR} = 10 \cdot \lg \left(\frac{M_1^2}{M_s} \right), \quad (6)$$

where M_1 represents the maximum possible pixel value, and M_s denotes the mean squared error.

SSIM is a perception-inspired metric that assesses structural similarity by comparing luminance, contrast, and structure between two images. It better aligns with human visual perception of detail and texture. The SSIM index is computed as:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (7)$$

where x and y represent the original and reconstructed images, respectively, μ_x, μ_y are their mean intensities, σ_x, σ_y are their standard deviations, σ_{xy} is their covariance, and C_1, C_2 are stabilization constants.

LPIPS measures perceptual similarity by leveraging features from a pre-trained deep neural network, effectively mimicking human judgment of image quality. It demonstrates high sensitivity to texture and detail distortion and possesses strong generalization capability, making it a leading metric for perceptual quality assessment. The LPIPS value is calculated as:

$$\text{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (x_{hw}^l - y_{hw}^l)\|_2^2, \quad (8)$$

where L is the total number of feature layers, l indexes the feature layer, H_l and W_l are the spatial dimensions (height and width) of the feature map at layer l , w_l is the channel-wise weight vector, and x_{hw}^l, y_{hw}^l are the feature representations of the original and reconstructed images at position (h, w) in layer l , respectively.

Higher PSNR and SSIM values, coupled with a lower LPIPS score, indicate superior reconstruction quality. Conversely, opposing trends suggest convergence difficulties during model training.

V. RESULTS

This section presents a systematic exposition of the experimental results and corresponding analyses. The evaluation follows a two-stage methodology: initially, we assess the performance of various edge detection algorithms to identify the optimal operator for image sharpness evaluation, subsequently, we integrate this selected operator into the 3D Gaussian

Splatting pipeline to validate its practical efficacy in enhancing reconstruction quality.

A. Edge Detection Performance Evaluation

TABLE I: SHARPNESS ASSESSMENT SCORES OF VARIOUS EDGE DETECTION ALGORITHMS FOR IMAGES WITH DIFFERENT BLUR TYPES

Blur Categories	Roberts Score	Prewitt Score	Sobel Score	Laplacian Score	Canny Score
Camera Shake Blur	55.5	58.8	58.0	49.3	28.8
Defocus Blur	53.9	56.4	55.3	51.2	29.6
Motion Blur	54.6	55.9	55.6	52.6	3.2
Normal	60.6	62.9	62.1	52.9	33

To systematically evaluate the performance differences of various edge detection algorithms on blurry images, this study compares five classical operators: Roberts, Prewitt, Sobel, Laplacian, and Canny. The weighted fusion score of edge density and edge continuity is adopted as a quantitative metric to assess the effectiveness of each algorithm in discriminating image sharpness. Experiments were conducted on a multi-view image dataset of strawberry plants, with images degraded by defocus, camera shake, and motion blur selected as test samples. The algorithm demonstrating the best overall performance was selected as the image sharpness detection operator for subsequent research.

The edge detection results are visually compared in Fig. 2. For clear images, the Canny operator extracts detailed, continuous, and well-localized edges, accurately delineating the contours of flowerpots and leaves. In blurry images, however, edges detected by Canny are significantly diminished, indicating its high sensitivity to image sharpness degradation. In contrast, first-order gradient operators (Roberts, Prewitt, and Sobel) detect numerous edges in clear images but retain a substantial number of spurious edges under blurry conditions, resulting in poor discriminability. The Laplacian operator preserves some contours in clear images but is highly susceptible to noise and blur, causing edges to be easily attenuated.

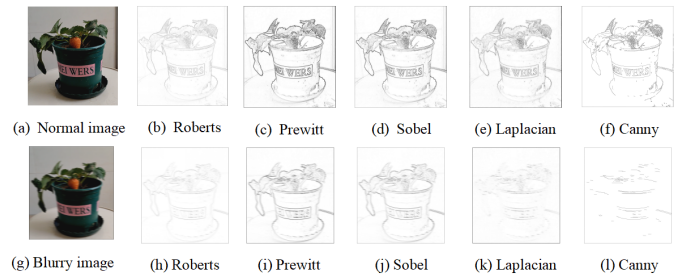


Fig. 2: Results of edge detection algorithms on normal and blurred images.

The superior performance of the Canny operator stems from its multi-stage processing pipeline, which includes Gaussian smoothing, gradient magnitude calculation, non-maximum

TABLE II: COMPARISON OF RECONSTRUCTION MODELS

Strawberry Growth Stages	NeRF			Instant-ngp			3DGS			3DGS-Canny		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Germination Stage	21.97	0.85	0.118	28.73	0.946	0.044	33.53	0.974	0.029	36.83	0.993	0.009
Flowering Stage	19.74	0.63	0.384	24.62	0.924	0.056	30.17	0.964	0.037	35.94	0.988	0.014
Fruiting Stage	18.81	0.58	0.412	23.18	0.902	0.067	29.72	0.963	0.038	34.87	0.982	0.024
Dormancy Stage	21.64	0.697	0.323	27.92	0.958	0.042	31.51	0.968	0.034	36.32	0.989	0.012
Mean	20.54	0.689	0.309	26.11	0.933	0.052	31.23	0.967	0.035	35.99	0.988	0.015

suppression, and dual-threshold edge linking. This comprehensive approach effectively suppresses noise-induced false detections while precisely preserving structural edges, thereby producing markedly distinct outputs for clear and blurred images. Table I presents the quantitative sharpness evaluation results for the five algorithms on images affected by different types of blur, based on the weighted fusion scores.

The experimental results demonstrate that the Canny operator significantly outperforms other algorithms in image sharpness assessment, with its superiority evidenced in two key aspects. First, it exhibits a wider dynamic range in output scores, indicating higher discriminatory sensitivity to sharpness variations. Second, it maintains optimal consistency and discriminative capability across different blur types, collectively confirming its sensitivity and robustness as a sharpness detection operator.

B. Evaluation of 3D Strawberry Plant Reconstruction

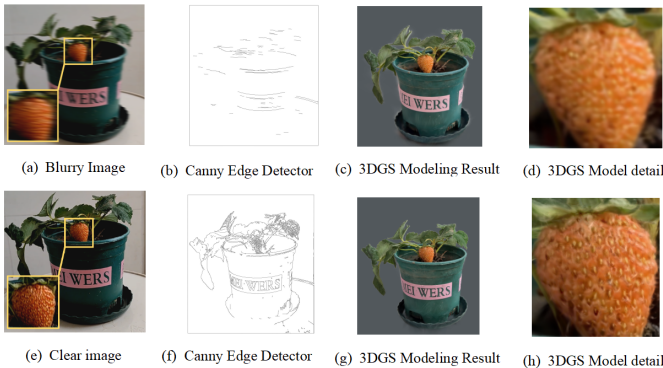


Fig. 3: Enhanced 3DGS reconstruction with integrated canny edge detection.

To validate the improvement in 3D Gaussian Splatting (3DGS) reconstruction accuracy achieved by integrating the Canny edge detection algorithm, a comparative experiment was conducted. The evaluation involved, first, a visual comparison of model details between the proposed Canny-3DGS framework and the native 3DGS to assess their performance in detail reconstruction. This was followed by a systematic, quantitative comparison of comprehensive reconstruction performance across different growth stages of strawberry plants, evaluating NeRF, Instant-ngp, native 3DGS, and the proposed Canny-3DGS fusion method.

The visual comparison results are shown in Fig. 3, confirming that input image quality significantly impacts the

reconstruction accuracy of 3DGS. When the dataset contains blurred images, the reconstructed model exhibits a substantial loss of microscopic geometric details on the fruit surface, manifested as blurred leaf vein edges and diffuse achene contours. The framework utilizing Canny for input filtering successfully mitigates this issue.

To quantitatively evaluate the performance of the different models, the PSNR, SSIM, and LPIPS values for each were calculated under identical training conditions. The detailed comparative results are provided in Table II.

The 3DGS-Canny framework achieved a remarkable improvement in reconstruction quality. Its average PSNR reached 35.99 dB, representing enhancements of 75.2%, 37.9%, and 15.% over NeRF, Instant-ngp, and native 3DGS, respectively. The average SSIM reached 0.988, showing improvements of 43.4%, 5.9%, and 2.2% compared to the three baseline methods. Meanwhile, the average LPIPS was reduced to 0.015, demonstrating reductions of 95.1%, 71.2%, and 57.1% relative to the three comparison methods. These results collectively validate the significant effectiveness of integrating Canny edge detection for enhancing the reconstruction accuracy of 3D Gaussian Splatting.

VI. CONCLUSION

This study introduces an enhanced 3D Gaussian splatting framework that integrates Canny edge detection technology, effectively suppressing plant reconstruction distortions caused by motion during image acquisition. By developing an image quality assessment mechanism based on edge features, the framework enables efficient screening of input images and significantly improves the reconstruction accuracy of strawberry plants. The proposed technical approach can be applied to crop phenotyping analysis and precision agriculture. Although satisfactory results have been achieved, the framework's performance in complex field environments requires further improvement. Future research will focus on extending this framework to additional crop species and exploring its integration with agricultural robots and unmanned aerial vehicles, with the goal of establishing a comprehensive plant phenotyping platform.

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